Section 12.5: Equations of Lines and Planes

Vectors and the Geometry of Space

MTH 310
Calculus III

Lines in 2D — Review

In 2D, a line is determined by a **point** and a **slope** (direction):

$$y = mx + b \qquad \text{or} \qquad y - y_0 = m(x - x_0)$$

Example 1

Find the equation of the line through $P(0,0)$ and $Q(1,2)$.

Find the slope, then use point-slope form.

Parametric Equations of Lines

Instead of $y = mx + b$, we can describe a line **parametrically**:

$$x = x_0 + at, \qquad y = y_0 + bt$$

The parameter t sweeps out the line. The **direction vector** $\vec{v} = \langle a, b \rangle$ tells which way the line goes.

Example 2

Write parametric equations for the line through $P(0,0)$ with direction $\langle 1, 2 \rangle$.

Use the parametric form with the given point and direction.

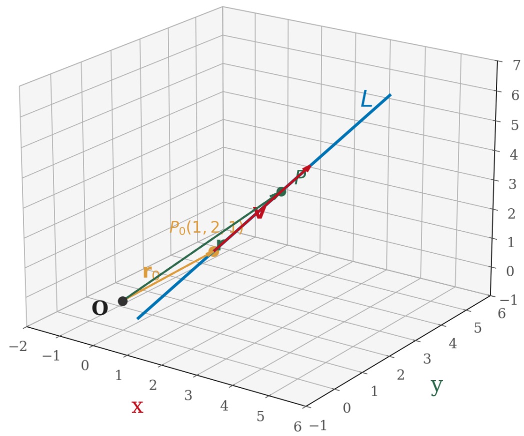
Vector Equation of a Line

Pack the parametric equations into one **vector equation**:

Definition 1: Vector Equation of a Line

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

- $\vec{r}_0$ = position vector of a known point on the line
- $\vec{v}$ = **direction vector** (parallel to the line)
- t = parameter (varies over all real numbers)



Desmos 3D [https://www.desmos.com/3d]

A line in **any dimension** needs just two things: a **point** and a **direction**.

Lines in 3D

The same formula works in 3D — no changes needed!

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

Example 3

Find vector and parametric equations for the line through $P(1, 1, 1)$ and $Q(4, 5, 6)$.

Find the direction vector PQ, then write the vector equation. Expand into parametric form.

Symmetric Equations

From the parametric equations, **solve each one for** t:

$$t = \frac{x-1}{3} = \frac{y-1}{4} = \frac{z-1}{5}$$

Definition 2: Symmetric Equations

If $\vec{v} = \langle a, b, c \rangle$ (all nonzero), the **symmetric equations** of the line through (x_0, y_0, z_0) are:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Uses:

- Quick way to check if a point is on the line (plug in and see if all three fractions are equal)
- Eliminates the parameter t

If one direction component is **zero**, that coordinate is constant — write it separately. For example, if $a = 0$:
$$x = x_0 \text{ and } \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Your Turn: Line Through Two Points

Example 4

Find parametric and symmetric equations for the line through $A(2, -1, 3)$ and $B(4, 3, 1)$.

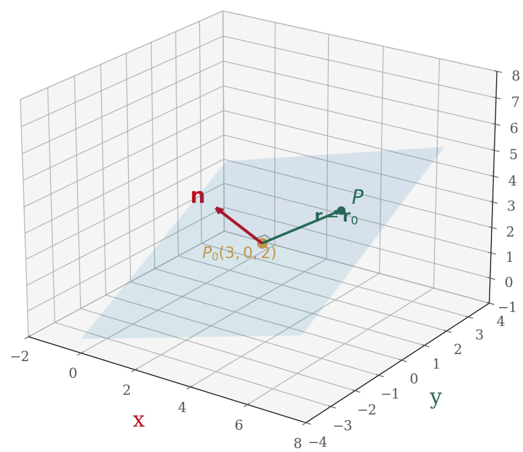
Find the direction vector AB, write parametric equations, then solve each for t.

Planes in 3D

A **plane** is determined by a point and a **normal vector** (perpendicular to the plane).

If $\vec{r}_0$ points to a known point P_0 and $\vec{n}$ is normal to the plane, then any point $\vec{r}$ in the plane satisfies:

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$$



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Definition 3: Equations of a Plane

- **Vector equation:** $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$
- **Scalar equation:** $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$, where $\vec{n} = \langle a, b, c \rangle$
- **Linear equation:** $ax + by + cz = d$

The coefficients of x, y, z in the plane equation $ax + by + cz = d$ **are** the components of the normal vector: $\vec{n} = \langle a, b, c \rangle$.

Example: Equation of a Plane

Example 5

Find the equation of the plane through $P(3, 0, 2)$ with normal vector $\vec{n} = \langle -1, -2, 3 \rangle$.

Use the scalar equation with the given point and normal vector.

Line-Plane Intersection

Example 6

Where does the line $\vec{r} = \langle 1, 2, -1 \rangle + t\langle 4, -1, 3 \rangle$ intersect the plane $-x - 2y + 3z = 3$?

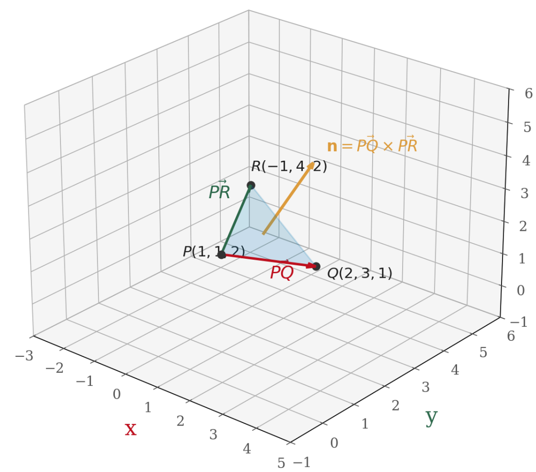
Substitute the parametric equations into the plane equation and solve for t.

Plane Through Three Points

Example 7

Find the equation of the plane through $P(1, 1, 2)$, $Q(2, 3, 1)$, and $R(-1, 4, 2)$.

Strategy: Find two vectors in the plane, then use the **cross product** to get the normal.



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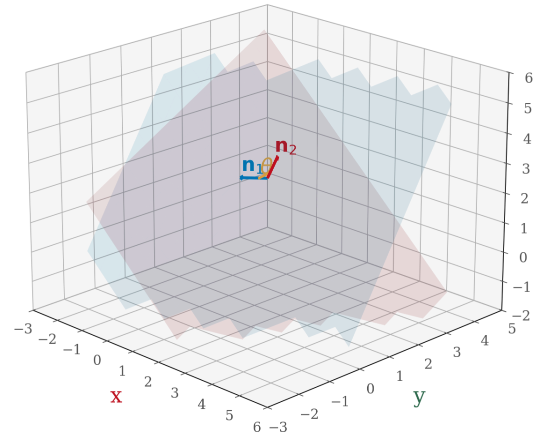
Find PQ and PR, compute their cross product, then use the result as the normal vector.

Angle Between Planes

The **angle between two planes** equals the angle between their normal vectors.



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Definition 4: Angle Between Planes

If two planes have normal vectors $\vec{n}_1$ and $\vec{n}_2$, the angle θ between the planes satisfies:

$$\cos\theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1|\ |\vec{n}_2|}$$

We use the **absolute value** of the dot product so that $0 \le \theta \le 90^\circ$.

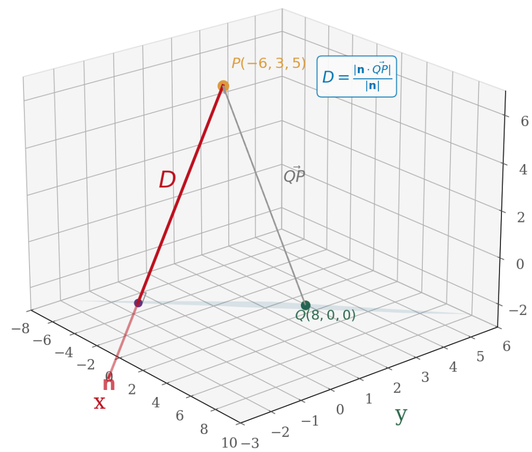
Example 8

Find the angle between the planes $x - 2y + z = 2$ and $3x - 2y + 3z = 3$.

Read off the normal vectors, then use the angle formula.

Distance from a Point to a Plane

To find the distance from a point to a plane, **project** a connecting vector onto the normal.



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Definition 5: Distance from a Point to a Plane

The distance from point $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz = d$ is:

$$D = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Example 9

Find the distance from $P(-6, 3, 5)$ to the plane $x - 2y - 4z = 8$.

Method: pick any point Q on the plane, form PQ, then project onto the normal. Or use the formula.

Geometry True/False — Lines and Planes in 3D

Let's test our spatial intuition! Decide whether each statement is **True** or **False**.

Statement	T/F
Two planes either intersect or are parallel.	
Two lines either intersect or are parallel.	
Two planes parallel to a third plane are parallel.	
Two planes orthogonal to a third plane are parallel.	
Two lines parallel to a plane are parallel.	
Two lines orthogonal to a plane are parallel.	
A plane and a line either intersect or are parallel.	

Think carefully about each one. Sketch examples and counterexamples.

Homework

Section 12.5: 2, 9, 10, 16, 21, 25, 30, 33, 57, 64, 71